

A Visibility Threshold for Top- k Logit-Lens Readouts

Olli Tuomi
Evident Solutions Oy

Abstract

Projecting a transformer hidden state, or a difference of hidden states, through the unembedding matrix W_U and reading the top- k tokens is a standard inspection technique. We ask how much of the state’s energy must be aligned with a token direction for that token to appear in the top k . Under an isotropy assumption on the remainder and on the rows of W_U , the top- k readout is a max-statistic and the answer is a closed form: $f^* = 2 \ln(2V)/(d + 2 \ln(2V))$, about 0.9% of the state’s energy for Phi-2 ($d=2560$, $V=51,200$). We test this with planted signals on ten models from five families (Pythia, Phi, GPT-2, BLOOM, Qwen, TinyLlama; $d = 512\text{--}2560$, $V = 32\text{k--}251\text{k}$). On the eight models whose unembedding passes the isotropy gate, the measured half-visibility point f_{50} tracks f^* with slope 1.15 in the log-log regression ($R^2 = 0.98$), with a family-dependent constant $f_{50}/f^* \in [0.56, 0.70]$ (0.56 ± 0.003 within the Pythia/Phi subset); the ceiling’s predicted rise with $\ln V$ is directionally confirmed across the $8\times$ vocabulary span. The two smallest Pythia models violate the assumption: their unembedding row directions are heavy-tailed (bath excess kurtosis $+1.1$ and $+2.2$ versus ≈ 0), the empirical ceiling is about twice the isotropic prediction, and the threshold rises accordingly (for Pythia-70m, a planted signal carrying 30% of the state’s energy is usually invisible). Averaging N independent difference vectors that share a signal lowers the threshold roughly N -fold, as predicted. Real hidden states violate the isotropy of the remainder as well (post-signal kurtosis 2.8–19.5), raising the effective bar for real reads by $1.05\text{--}1.5\times$ on the gate-passing models. This is a measured correction, not a failure of the framework. A battery of eight real contrast pairs clears the bar at $4.5\text{--}16\times$ on every isotropic model (24/24 present, surviving the real-remainder correction) and is absent on both anisotropic models (16/16). The practical readings are: a clean top- k readout is compatible with the large majority of the state’s energy being unaligned with any token direction; absence from the top k bounds aligned energy only up to f^* ; and the threshold is a property of W_U geometry, not of d alone, so small-model readouts require a measured calibration.

1 Introduction

The logit lens (nostalgebraist, 2020) and its descendants project hidden states through the unembedding matrix $W_U \in \mathbb{R}^{V \times d}$ (we write $\hat{W}_U$ for its row-normalised version) and read the top- k tokens. The same readout is applied to weight vectors such as MLP output columns (Geva et al., 2022; Dar et al., 2022). Differences of hidden states are not new either: contrastive and steering methods (Zou et al., 2023; Turner et al., 2023; Rinsky et al., 2024) construct them, though they interpret them through scalar projections or effects on generation rather than token readouts; Tuomi (2026) reads such differences through the logit lens, where the shared component cancels and the distinguishing tokens surface. In each reading use, a short list of tokens is taken as a description of what a d -dimensional vector contains.

A basic quantitative question about this readout appears not to have a standard answer: how large must a token-aligned component be, relative to everything else in the vector, before it surfaces in the top k ? The question matters in both directions. If the answer is “small,” then a clean token list is compatible with a vector whose energy is almost entirely elsewhere, and the readout should not be treated as an energy

account. If a component below the answer is present, the readout will not show it, and its absence should not be treated as evidence of absence.

This paper derives the threshold under explicit assumptions (§2), measures it with planted signals on five models (§3, §4), characterises where and why the assumptions fail (§4), and places one real contrast on the resulting curve at each scale (§5). All experiments are weights-plus-forward-pass measurements on public checkpoints; scripts, seeds, and result files are listed in §9.

2 The threshold

Setup. Let $x \in \mathbb{R}^d$ be the vector to be read (a hidden state, a difference of hidden states, or a weight column). Write it as

$$x = \sqrt{f} \|x\| u_t + \sqrt{1-f} \|x\| \varepsilon,$$

where u_t is the unit direction of the W_U row for token t , f is the fraction of x 's energy aligned with u_t , and ε is a unit vector. The readout computes $|\hat{W}_U x|$ for the row-normalised unembedding $\hat{W}_U \in \mathbb{R}^{V \times d}$ and returns the k tokens with the largest values. Token t is *visible* if it is among them.

Assumptions. (i) ε is isotropic; (ii) the rows of $\hat{W}_U$ behave, against a random direction, like V independent isotropic directions; (iii) the bar is set by the maximum bath entry ($k=1$); a top- k criterion replaces the maximum by the k -th order statistic and relaxes the bar by a constant factor. Assumption (ii) is the one that fails in practice, in a measurable way (§4).

Derivation. Under (i)–(ii), the bath entries $\hat{W}_U \varepsilon$ are approximately $\mathcal{N}(0, 1/d)$, and the maximum of V two-sided Gaussian draws concentrates at $\sqrt{2 \ln(2V)}$ standard deviations. Token t 's entry is $\sqrt{f} \|x\|$ against a bath of scale $\sqrt{(1-f)/d} \|x\|$. Visibility requires

$$\frac{\sqrt{f}}{\sqrt{1-f}} > \frac{\sqrt{2 \ln(2V)}}{\sqrt{d}} \iff f > f^* = \frac{2 \ln(2V)}{d + 2 \ln(2V)}.$$

For Phi-2 ($d=2560$, $V=51,200$): $f^* \approx 0.89\%$ and the bath ceiling is 4.80σ . Since $d \gg \ln V$ in practice, $f^* \approx 2 \ln(2V)/d$: the *energy* threshold is linear in $1/d$ (the *amplitude* bar $\sqrt{f^*}$ scales as $1/\sqrt{d}$). Because the dependence on V is logarithmic and V is nearly constant across the model suite used below, f^* is in practice a function of d .

Two consequences follow directly. First, a vector can carry, say, 90% of its energy in directions unaligned with every token and still produce a stable, clean top- k list from the remaining 10%, because $10\% \gg f^*$. Second, averaging N independent vectors that share the aligned component suppresses the incoherent remainder by $\sqrt{N}$ in amplitude, so a signal at $f < f^*$ becomes visible at roughly $N \approx f^*/f$.

3 Planted-signal measurements on Phi-2

Method. A target row direction is planted at energy fraction f into an isotropic bath, $\delta = \sqrt{1-f} \varepsilon + \sqrt{f} u_t$, and we record whether t lands in the top 15 of $|\hat{W}_U \delta|$. 100 samples per f ; content-word tokens; seed fixed. The $f=0$ bath maximum measures the ceiling directly: 4.46σ against the predicted 4.80σ , a $\sim 7\%$ undershoot attributable to correlations among W_U rows.

model	d	bath (σ)	analytic f^*	f_{50}	f_{50}/f^*
Pythia-70m	512	9.37	4.31%	no crossing	—
Pythia-160m	768	8.34	2.91%	14.70%	5.05
Pythia-410m	1024	4.41	2.20%	1.24%	0.561
Pythia-1b	2048	4.49	1.11%	0.62%	0.558
Phi-2	2560	4.46	0.89%	0.50%	0.561

Table 1: Planted-signal sweep across five models. f_{50} is the planted-energy fraction at which the top-15 hit rate crosses 0.5 (log-linear interpolation). The predicted ceiling is 4.80σ at every d ; the first two models fail the isotropy gate and are excluded from the regression.

Threshold curve. Hit rates for a single planted token:

f	0.001	0.003	0.01	0.03	0.1	0.3
hit rate	0.01	0.14	0.90	1.00	1.00	1.00
median rank	4119	307	0	0	0	0

The transition brackets $f^* \approx 0.9\%$. A 5-token family signal (the planted energy split over five related rows) shifts the curve right by roughly the per-token mass split, as expected: hit rate 0.48 at $f=0.01$, 0.91 at $f=0.03$.

Triangulation. Averaging N independent baths sharing a planted signal at $f = 0.003$ (below threshold):

N	1	2	4	8	16
hit rate	0.25	0.62	0.94	1.00	1.00
median rank	135	4	0	0	0

The predicted rescue point is $N \approx f^*/f = 3$; the curve crosses between $N=2$ and $N=4$. This is the arithmetic behind multi-contrast averaging (Tuomi, 2026): it does not sharpen the signal, it suppresses the bath.

4 The threshold across scale

V is fixed across the Pythia suite (50,277 after removing padded rows) and d spans $5\times$, so the closed form predicts f^* moves $4.8\times$. The experiment was pre-registered with three outcome branches (slope -1 : the closed form holds; slope -0.5 : the energy/amplitude exponent is wrong; slope 0 : the max-statistic model does not govern) and three gates, of which one (the bath ceiling must be roughly constant in d , since it depends only on V) doubles as a per-model isotropy check. A model failing it is reported and excluded from the regression. 200 samples per point.

Result on the isotropic subset. For the three models whose bath sits at the predicted ceiling (spread 0.08σ), regressing $\log f_{50}$ on $\log d$ gives slope -0.988 ($R^2 = 1.000$, $n=3$). More informative than the slope is the ratio column of Table 1: $f_{50}/f^* = 0.56$ at all three values of d , with spread 0.003. The closed form is correct up to a single multiplicative constant. The constant being below 1 has the expected sign: a top-15 hit needs to beat the 15th order statistic rather than the maximum, and the measured ceiling itself

model	d	V	tied	bath (σ)	f^*	f_{50}	f_{50}/f^*
Qwen2.5-0.5B	896	151,665	yes	4.76	2.74%	1.76%	0.64
GPT-2-medium	1024	50,257	yes	4.67	2.20%	1.54%	0.70
BLOOM-560m	1024	250,680	yes	4.79	2.50%	1.73%	0.69
Qwen2.5-1.5B	1536	151,665	yes	4.81	1.62%	0.92%	0.57
TinyLlama-1.1B	2048	32,000	no	4.41	1.07%	0.63%	0.59

Table 2: The sweep on five additional families. All pass the bath-ceiling gate; predicted ceilings vary with V (§4, V -dependence).

sits $\sim 7\%$ under the analytic value. We have not derived it; the scale-invariance of the ratio is the measured content.

Two known effects account for most of the constant. Replacing the maximum by the k -th order statistic moves the bar from $\sqrt{2 \ln(2V)}$ to approximately $\sqrt{2 \ln(2V/k)}$, an energy factor $\ln(2V/k)/\ln(2V) \approx 0.77$ at $k=15$, $V \approx 50k$; and the bar is set by the *measured* ceiling rather than the analytic one, a further factor $(4.46/4.80)^2 \approx 0.86$. Together these predict a ratio of ≈ 0.66 against the measured 0.56; we attribute the residual ≈ 0.85 factor to fluctuation of the signal entry itself at the 50% crossing (the planted row’s logit carries its own share of the bath noise) and to residual row correlations, and present this as intuition rather than derivation. The decomposition does predict that the constant should drift with k and with the measured ceiling, which is testable.

Five more families. To move past an $n=3$ regression, we ran the identical sweep on five further models chosen to break the suite’s shared tokenizer and to span vocabulary size: Table 2. All five pass the bath-ceiling gate, giving eight isotropic models across five families, $d = 896$ –2560 and $V = 32k$ –251k. Because V now varies, the right regression is $\log f_{50}$ on $\log f^*$ (which absorbs both d and V): slope 1.15, $R^2 = 0.981$, $n=8$. The order-statistic constant is family-dependent (0.56 for Pythia/Phi, 0.57–0.70 for the others), which is why the mixed-family slope sits above 1; the within-family result (slope -0.99 in $\log d$, constant 0.56) remains the cleanest statement, and the family spread bounds how far the constant travels: every measured value lies in $[0.56, 0.70]$.

The V -dependence, first contact. The closed form predicts the ceiling rises as $\sqrt{2 \ln(2V)}$: 4.71σ at $V=32k$, 4.80σ at $50k$, 5.02σ at $152k$, 5.12σ at $251k$. Across the eight gate-passing models the measured ceilings correlate with $\ln V$ at 0.88 and rise monotonically with V within matched d (BLOOM-560m 4.79σ vs. GPT-2-medium 4.67σ at $d=1024$), sitting uniformly 4–7% under the analytic value, the same undershoot as Phi-2’s and attributable to row correlations. The V -term is directionally confirmed over an $8\times$ span; a quantitative fit would need more models per V .

Where the assumption fails. Pythia-70m and 160m violate the ceiling gate at 9.37σ and 8.34σ , roughly double the isotropic prediction. The consequence for readability is large: 160m’s measured f_{50} is 14.7%, five times its analytic f^* , and 70m’s hit rate is only 0.38 at $f = 0.3$: a signal carrying 30% of the state’s energy usually loses the top-15 to the bath.

A follow-up diagnostic (not pre-registered) checked the obvious explanation, axis-aligned outlier coordinates (Timkey and van Schijndel, 2021; Puccetti et al., 2022): it does not hold for this object. On the row-normalised W_U , per-coordinate energy concentration is mild in all five models (top-1 coordinate share 0.0015–0.0065), and removing the eight highest-variance coordinates leaves the inflated ceilings unchanged (70m: $9.37\sigma \rightarrow 9.61\sigma$; 160m: $8.60\sigma \rightarrow 8.55\sigma$; the diagnostic re-draws the bath, so its 160m

model	layer	bar f_{emp}	f_{real}	margin	signal
Pythia-70m	4/6	14.64%	1.97%	—	absent ($0.5\times$ null)
Pythia-160m	8/12	8.78%	1.27%	—	absent ($0.6\times$ null)
Pythia-410m	15/24	1.91%	17.32%	$9.1\times$	present
Pythia-1b	10/16	0.93%	8.90%	$9.5\times$	present
Phi-2	20/32	0.77%	9.23%	$12.1\times$	present

Table 3: One real contrast placed on each model’s curve. Margin is $f_{\text{real}}/f_{\text{emp}}$.

baseline differs from the sweep’s 8.34σ by sampling). What separates the failing models is the excess kurtosis of the bath distribution: $+1.14$ (70m) and $+2.15$ (160m) versus 0.00 – 0.01 for the three passing models. The small models’ unembedding *row directions* are heavy-tailed in a way not carried by any small set of coordinates; we leave the precise geometry open. Operationally: the isotropic threshold formula applies exactly where the bath is Gaussian-tailed, and the bath kurtosis is a one-line check for whether it applies.

5 A real contrast on the curve, at each scale

The planted-signal experiments locate the bar; they say nothing about what real vectors carry, and they construct the isotropic remainder of assumption (i) by design. We therefore first measure that assumption on real vectors, then place a battery of real contrasts on the curve.

Assumption (i) on real vectors. For the three isotropic models at the capture layers below, we take real hidden states, remove the span of their top-15 W_U rows (the “signal”), and measure the remainder’s bath statistics through $\hat{W}_U$. The remainder is *not* isotropic: median excess kurtosis is 2.8 (Pythia-410m), 19.5 (Pythia-1b), 3.6 (Phi-2), with ceilings of 4.9σ , 6.6σ , 5.5σ against the 4.8σ prediction. The same statistic on the real contrast *difference* below gives kurtosis $4.0/6.2/8.4$ and ceilings $4.5\sigma/5.2\sigma/5.5\sigma$. The practical consequence is a multiplicative correction: using each model’s real-difference ceiling in place of the random-bath ceiling scales the bar by $(s_{\text{real}}/s_{\text{rand}})^2 = 1.05$ (410m), 1.32 (1b), 1.50 (Phi-2). The framework survives with a measured correction factor; the margins reported below clear it comfortably.

The anchor. As a real anchor we use a minimal contrast pair inspired by prior work with this readout (Tuomi, 2026): the difference of hidden states for “My grandmother said that the {hot, cold} dog that she saw was” at the final position, captured at the layer nearest 62.5% depth in each model. The aligned-energy fraction f_{real} is the least-squares span energy of the top-15 W_U rows on the difference vector; the null is the median span energy of 15 random rows. The per-model bar is the empirical $f_{\text{emp}} = s^2/(d + s^2)$ computed from that model’s measured bath ceiling s , which is the calibrated bar for the two anisotropic models. A model’s margin is interpreted only if $f_{\text{real}} \geq 3\times$ its random-span null; below that we report signal absence rather than a margin.

On the three isotropic models the margin is 9 – $12\times$ the bar and roughly constant: as d falls from 2560 to 1024 the bar rises $2.5\times$ and the contrast’s aligned energy rises with it ($9.2\% \rightarrow 17.3\%$). For this contrast, readability does not degrade with d in the isotropic regime: bar and signal move together. The top-15 lists are recognisably about the contrast at all three scales (e.g., Pythia-1b’s top-15 includes *salad*, *ingredients*, *creatures*, *breeds*).

On the two anisotropic models the contrast is signal-absent: the top-15 consists of whitespace and byte tokens, and the span energy sits below even the random-row null (the selected junk tokens’ rows are

nearly collinear, so fifteen of them span less than fifteen random rows). This grouping is expected rather than mysterious: rare and undertrained tokens retain unembedding rows near their common initialisation cluster, so a heavy-tailed bath direction that overlaps the cluster promotes the whole group at once: the readout-side signature of the undertrained-token phenomenon. With a single contrast we cannot distinguish “the model never forms the distinction” from “the distinction is formed and the heavy-tailed bath swamps it”; both are consistent with the observation. We note only that the two models where the readout geometry is degraded are also the two where the one real read we have is empty.

A battery, not an anecdote. To test whether the anchor is typical, we ran seven further minimal pairs (soup temperature, food/tool, animal/vehicle, negation, king/queen, summer/winter, coffee/water) through the same machinery on all five models. On the three isotropic models every pair is signal-present (24/24), with median aligned energy $13.0\%/8.2\%/9.2\%$ and median margins $6.8\times/8.8\times/12.0\times$ (range $4.5\text{--}16\times$); the hot/cold anchor sits near each model’s median, not above it. On the two anisotropic models every pair is signal-absent (16/16), extending the single-anchor observation: where the readout geometry is degraded, none of eight ordinary contrasts survives the bath. These margins are quoted against the random-bath bar; applying the real-difference correction above ($1.05\text{--}1.5\times$) leaves all 24 present.

This is still one read position and one seed per pair; the battery anchors the synthetic curves in eight real measurements per scale rather than one, but it is not a population claim about what models of different sizes carry.

6 What the threshold does and does not license

Clean readouts are cheap. Above f^* (about 1% of energy at $d \approx 2500$), a token direction dominates the top- k list. A stable, interpretable readout is therefore compatible with $\geq 90\%$ of the vector’s energy being unaligned with every token direction, and a top- k list should not be read as an energy account of the state.

Absence is only bounded. A token missing from the top k carries less than roughly f^* aligned energy at that layer (up to the order-statistic constant); it is not evidence of zero signal. Averaging N matched vectors lowers the detection bar to about f^*/N .

The bar is W_U -geometry, not d . The isotropic formula holds (up to the family constant) on eight of ten models across five families, including one at $d=896$; the two failures are the smallest Pythias, and what separates them is bath kurtosis, not dimension. Readout-based claims on small models should be calibrated against the measured bath of that model’s W_U , not the dimension count; claims about real states should apply the measured real-remainder correction (§5).

7 Related work

The logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) established the readout studied here; Geva et al. (2022) and Dar et al. (2022) apply it to weight vectors. Contrastive and steering methods (Zou et al., 2023; Turner et al., 2023; Rimskey et al., 2024) construct differences of hidden states but interpret them through scalar projections or behavioural effects, not token readouts; Tuomi (2026) reads such differences through the logit lens, where the shared component cancels and the threshold derived here governs what surfaces from the remainder. Anisotropy of learned embedding spaces is well documented (Ethayarajh, 2019; Timkey and van Schijndel, 2021; Puccetti et al., 2022), including massive activations in the residual stream (Sun et al., 2024); the measurement here isolates the specific consequence of unembedding row-direction anisotropy for top- k readability, and finds it is not the axis-aligned form in the two affected models. Heap et al. (2025) show sparse-autoencoder interpretability metrics can score well

on random-weight models; the threshold arithmetic is a possible ingredient of such effects (any readout that only needs $\sim 1\%$ of energy aligned will find something to read in many vectors), though we have not tested that connection.

8 Limitations

- The derivation assumes an isotropic remainder and isotropic row directions. The second fails measurably on two of ten models; the first fails on real hidden states (post-signal kurtosis 2.8–19.5), where it is handled as a measured $1.05\text{--}1.5\times$ bar correction rather than derived.
- The within-family regression has $n=3$ points over a $2.5\times$ span of d ; the mixed-family regression has $n=8$ but conflates the family-dependent constant with the slope.
- The order-statistic constant (f_{50}/f^* ; 0.56 for Pythia/Phi, 0.56–0.70 across families) is measured, not derived, will vary with k , and its family dependence is unexplained.
- The battery uses one read position and one seed per pair; the signal-absent results on the two small models are confounded by model competence and are reported as absences, not as readability failures.
- Row-normalised readout ($|\hat{W}_U x|$) is used throughout; the unnormalised logit-lens variant re-weights ranks by row norms and its constant may differ.
- All models are $\leq 2.7\text{B}$ parameters. The V -dependence of the ceiling is confirmed only directionally ($V = 32\text{k--}251\text{k}$, correlation 0.88, uniformly 4–7% under the analytic value); the V -dependence of f_{50} itself is untested per-family.

9 Reproducibility

All experiments run on public checkpoints (EleutherAI Pythia deduped suite; microsoft/phi-2; gpt2-medium; bigscience/bloom-560m; Qwen/Qwen2.5-0.5B and -1.5B; TinyLlama/TinyLlama_v1.1) with fixed seed 0. Scripts and result files: `census_visibility_threshold.py` (Phi-2 curves, triangulation, real anchor), `census_visibility_threshold_scale.py` (five-model sweep, gates, regression), `census_visibility_threshold_scale_posthoc.py` (kurtosis diagnostic), `census_visibility` (real anchor per scale), and `census_visibility_revision.py` (five additional families, the real-vector isotropy check, and the eight-pair battery), each writing a JSON artifact with full per-point numbers.

Use of AI Assistants

Large language models were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the authors, who take full responsibility for the content.

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